

Ryan Cho

Senior Software Engineer | ML Systems & Infrastructure

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PROFESSIONAL SUMMARY

Software engineer with 7+ years delivering production systems across investment management and financial services. Focuses on Python ML pipelines, scalable data workflows, and model serving infrastructure that shortens research iteration cycles. Hands-on with PyTorch and TensorFlow training pipelines, GPU-accelerated workloads, and real-time inference integration. Partner to researchers and engineering leaders on reliability, observability, and production readiness at platform scale.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected **Python-based ML training pipelines** supporting batch and scheduled experiment runs, reducing researcher turnaround from days to hours across **40+ model iterations** per release cycle.
- Built **data curation and preprocessing workflows** for structured and time-series datasets at scale, enabling reliable ingestion for **PyTorch** and **TensorFlow** training jobs.
- Designed **model integration systems** connecting research checkpoints to production inference services with versioning, rollback, and automated validation gates.
- Developed **GPU-accelerated distributed training** infrastructure on AWS, improving throughput for large-scale experiments while maintaining cost visibility through resource tagging and quotas.
- Implemented **MLOps tooling** including experiment tracking, metadata management, and pipeline observability using MLflow and custom dashboards for cross-team visibility.
- Optimized **real-time inference endpoints** for latency-sensitive analytics, achieving sub-100ms p95 response through batching, caching, and model graph optimization.
- Collaborated with **quantitative researchers** to translate experimental prototypes into maintainable, tested production code with CI/CD coverage and structured code review standards.
- Established **data collection pipelines** with automated quality checks and lineage tracking, supporting reproducible datasets for longitudinal model evaluation.
- Contributed to shared infrastructure and **cross-functional tooling**, including internal SDKs that simplified experiment deployment for adjacent engineering teams.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Built **Python and Java services** for risk analytics pipelines processing daily market datasets across **12 upstream feeds** with automated reconciliation.
- Developed **ETL workflows** and batch data pipelines on internal research platforms, improving data freshness SLA compliance to **99.5%** quarterly.
- Implemented **model deployment infrastructure** for quantitative research tools, enabling controlled promotion from sandbox to production with audit trails.
- Designed **REST and messaging integrations** connecting trading analytics modules with enterprise data warehouses and reporting systems.
- Created **automation tooling** reducing manual data preparation steps for analysts, saving an estimated **15 hours per week** across the team.
- Participated in **code reviews** and engineering standards initiatives, raising test coverage and documentation quality for shared libraries.
- Collaborated with **stakeholders and quants** to translate research requirements into reliable, performant backend systems under regulatory constraints.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed **full-stack features** for inventory and compliance web applications using React and Node.js, supporting **3 production releases** during the internship.
- Built **REST APIs** and database integrations enabling real-time check-in workflows for restaurant partners across New York metro locations.
- Implemented **automated testing** and CI pipelines improving deployment confidence for weekly release cadences.
- Contributed to **data export and reporting tools** helping operations teams track usage metrics and compliance status.
- Partnered with senior engineers on **code reviews** and refactoring initiatives, adopting team conventions for maintainable service architecture.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web interfaces** for nonprofit fundraising campaigns using HTML, CSS, and JavaScript, supporting launch of **2 client sites** on schedule.
- Integrated **third-party APIs** for donation processing and email outreach, ensuring secure handling of donor data.
- Collaborated with designers to implement **accessible UI components** aligned with WCAG guidelines across public-facing pages.
- Maintained **content management workflows** enabling non-technical staff to update campaign materials without engineering support.
- Documented **deployment procedures** and handoff materials for continuity after project completion.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Responsive Design

Backend: Python, Java, Go, FastAPI, Node.js, REST APIs, gRPC, Distributed Systems

Databases: PostgreSQL, MongoDB, Redis, Snowflake, Data Lakes, ETL Pipelines

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, GPU Computing

Monitoring & Tools: Prometheus, Grafana, MLflow, Weights & Biases, PyTorch, TensorFlow, Git

Workflow & PM: Agile, Scrum, Jira, Confluence, Technical Documentation, Cross-Team Collaboration